

KRISHEN MATHUR

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Education

University of Illinois at Urbana-Champaign

Expected December 2027

B.S. Electrical Engineering, Minor in Semiconductor Engineering GPA: 3.50

Champaign, IL

- **Relevant Coursework:** Analog Signal Processing, Digital/Analog Circuits, Semiconductor Electronics, Computer-Aided Design (PCB), Computer Systems and Programming, Probability and Statistics

Projects

Analog DJ Mixing Board (*LTspice, Altium, Python*)

May 2026 – Present

Personal Project | Analog Circuit Design, PCB Design

- Designing a 2-channel analog DJ mixer with op-amp gain stages, summing amplifier, 3-band EQ, and crossfader modules, progressing from breadboard prototype to fabricated PCB.
- Laying out a multi-layer PCB in **Altium** with dedicated ground planes and power decoupling to minimize channel crosstalk and noise coupling.
- Characterizing each module on the bench with oscilloscope, function generator, and DMM, benchmarking measured frequency response, gain, THD, and noise floor against LTspice-simulated targets.
- Built a **Python** workflow to process bench measurements, generate Bode plots, and automate measured-vs-simulated response comparison.

Project Stealth – Wearable EMG & IMU System (*Arduino, C++, Python, BLE*)

Aug 2025 – Present

I-MADE (*Illinois Medical Advancements through Design and Engineering*)

Champaign, IL

- Engineered a 2-channel wearable EMG + IMU acquisition system (Arduino Nano ESP32, MyoWare 2.0, MPU6050) streaming synchronized biosignal data over **BLE** to a live Python pipeline.
- Designed the signal-conditioning chain with instrumentation amplification and 20–450 Hz active band-pass filtering to isolate the surface-EMG band for high-fidelity capture.
- Trained an SVM classifier on filtered EMG features to detect muscle activation in real time, driving a live gesture-controlled game demo for 300+ attendees at Engineering Open House.
- Pitched the system at the Cozad New Venture Challenge; wrote embedded firmware for multi-channel sampling, power optimization, and CSV logging for offline analysis.

Work Experience

Power Electronics Intern – Sourcing & Development (*LTspice, EBOM, Power Electronics*)

Summer 2026

Centre for Innovation, Technology and Learning (CITL) – Hero MotoCorp

Jaipur, India

- Simulated a custom active PFC boost converter in **LTspice** for a 580W Vida onboard charger, achieving **94% simulated efficiency** using two-loop average current mode control.
- Built a ~120 line-item EBOM spanning power MOSFETs, magnetics, connectors, potting, and enclosure, supporting sourcing decisions and vendor negotiations for OBC production.
- Authored a concept proposal for a 580W MagSafe-style wireless EV charging system, evaluating coil coupling, alignment, and thermal constraints for two-wheeler EVs.
- Analyzed the complete OBC architecture (EMI filtering, rectification, PFC, LLC resonant stage, control) and audited SMT/assembly lines to map end-to-end charger production.

Team Lead, Student Technology Consultants (*Leadership, Networking, AV Systems*)

Mar 2025 – Present

University of Illinois at Urbana-Champaign – Technology Services

Champaign, IL

- Lead a team of **30+ consultants** handling **500–800 support tickets per week**; coordinate coverage and escalations to sustain daily campus-wide IT operations.
- Train new hires on diagnostic workflows, ticketing procedures, and university systems, accelerating ramp-up and improving first-contact resolution.
- Resolve complex escalated tickets across networking, account access, and hardware, and manage AV/institutional technology across classroom and event spaces.

Technical Skills

Hardware & Circuits: Analog Circuit Design, PCB Design (Altium), Op-Amp Circuits, Signal Conditioning, BMS, Soldering, Breadboarding

Measurement & Test: Oscilloscope, Function Generator, DMM, Frequency Response Characterization (Bode, THD, Noise Floor), SPICE Simulation, Hardware Debugging

Programming: Python (data analysis, signal processing), C, C++, Assembly, Arduino, Embedded Firmware, Serial Communication, BLE

Software & Tools: Git/GitHub, Autodesk Fusion 360, LaTeX, Linux

Other: Technical Communication, Public Speaking, Team Leadership

Interests: Strength Training, Electronic Music, Football, Billiards, Table Tennis